

Customer shopping behaviour: what the data says

Repeat behaviour, revenue concentration, and the signals that point to action.

The customer base is already repeat-led

3,116

Loyal

customers

701

Returning

customers

83

New

customers

CUSTOMERS BY SEGMENT

Segment



SO WHAT

The growth opportunity is less about first purchase and more about deepening repeat behaviour.

A clean pipeline turned raw transactions into decisions

01

Clean

Clean and standardize the dataset in Python.

3,900

purchases across product categories

02

Engineer

Engineer age groups and purchase-frequency features.

18

columns spanning demographics, products, promotions, reviews, and shipping

03

Structure

Load the prepared data into MySQL for structured analysis.

37

missing review ratings, imputed with category medians

04

Decide

Surface the findings in Power BI.

SO WHAT

The dataset is large enough to reveal behaviour patterns.

Revenue is concentrated in the non-subscriber base

Subscription is not yet a higher-spend engine, but it is a clear conversion opportunity.

\$170,436

Non-subscribers

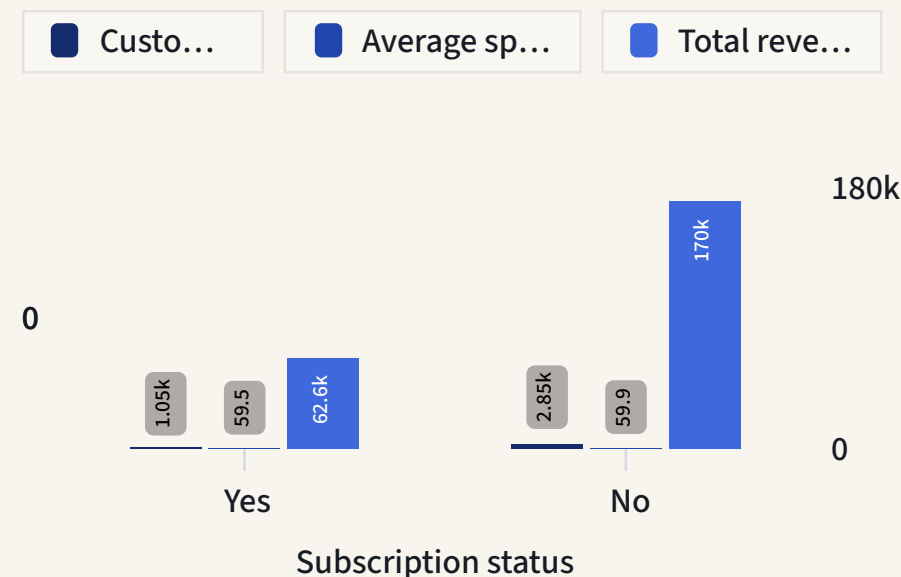
2,847 customers · \$59.87 average spend

\$62,645

Subscribers

1,053 customers · \$59.49 average spend

REVENUE AND AVERAGE SPEND BY SUBSCRIPTION STATUS



SO WHAT

Convert the larger non-subscriber base without assuming subscription already lifts spend.

Repeat behaviour is the strongest retention signal

Customers are segmented by purchase history into New, Returning, and Loyal groups.



New

First purchase



Returning

Purchase history
builds



Loyal

Repeat behaviour
compounds

958

subscribers among
customers with more
than five purchases

2,518

non-subscribers
among customers
with more than five
purchases

SO WHAT

Identify repeat buyers, then give them a reason to subscribe.

Small delivery upgrades correlate with higher basket value

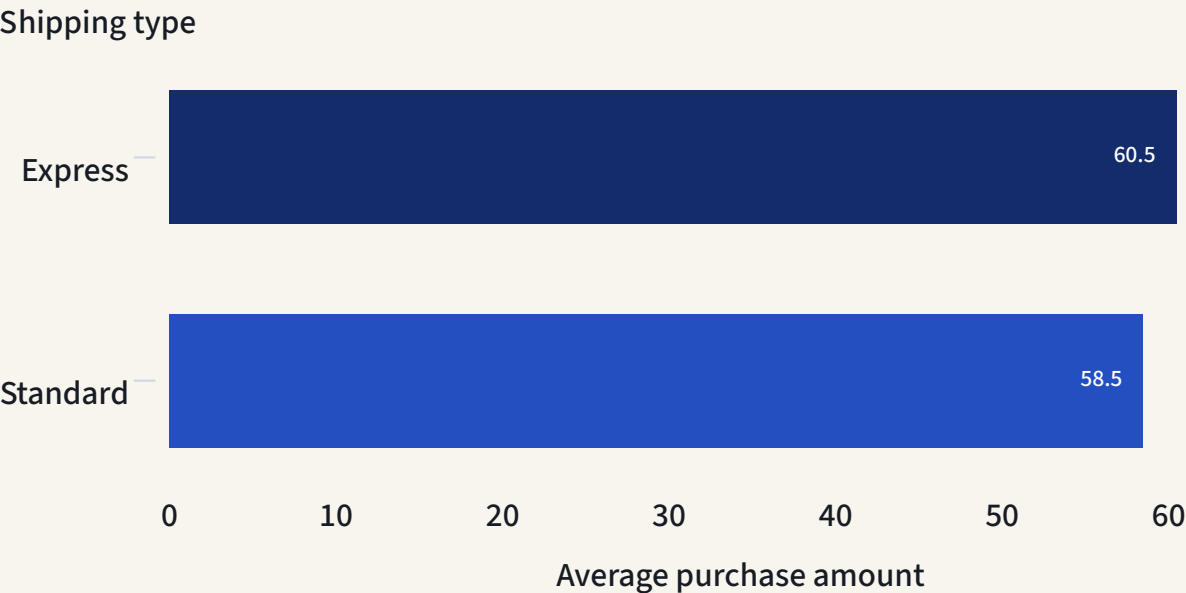
\$2.02

directional gap

Express-shipping orders have a modestly higher average purchase amount than standard-shipping orders.

Use the gap to test shipping-led upsells and premium service bundles.

AVERAGE PURCHASE AMOUNT BY SHIPPING TYPE



SO WHAT

Test shipping-led upsells and premium service bundles.

Product signals split between quality and promotion

Some products earn attention through ratings. Others rely more heavily on discounts.

Signal	Products
Top-rated	Sandals 3.82; Boots 3.79; Gloves, Hat, and Handbag at 3.78
Most discount-dependent	Hat 50.00%; Sneakers 49.66%; Coat 49.07%; Sweater 48.17%; Pants 47.37%

SO WHAT

Protect margin on promotion-dependent products while featuring quality winners.

Best sellers give campaigns a ready-made product hierarchy

Turn the category leaders into a campaign matrix rather than a flat product list.

Category	Lead product	Next products	Campaign role
Accessories	Jewelry • 171 orders	Sunglasses and Belt • 161 each	Hero, then supporting accessories
Clothing	Blouse and Pants • 171	Shirt • 169	Pair the leaders in the opening frame
Footwear	Sandals • 160	Shoes • 150; Sneakers • 145	Lead with the category anchor
Outerwear	Jacket • 163	Coat • 161	Use a tight two-product story

SO WHAT

Feature top-rated and best-selling products in campaigns, category by category.

Young adults are the largest revenue-contributing age group

\$62,143

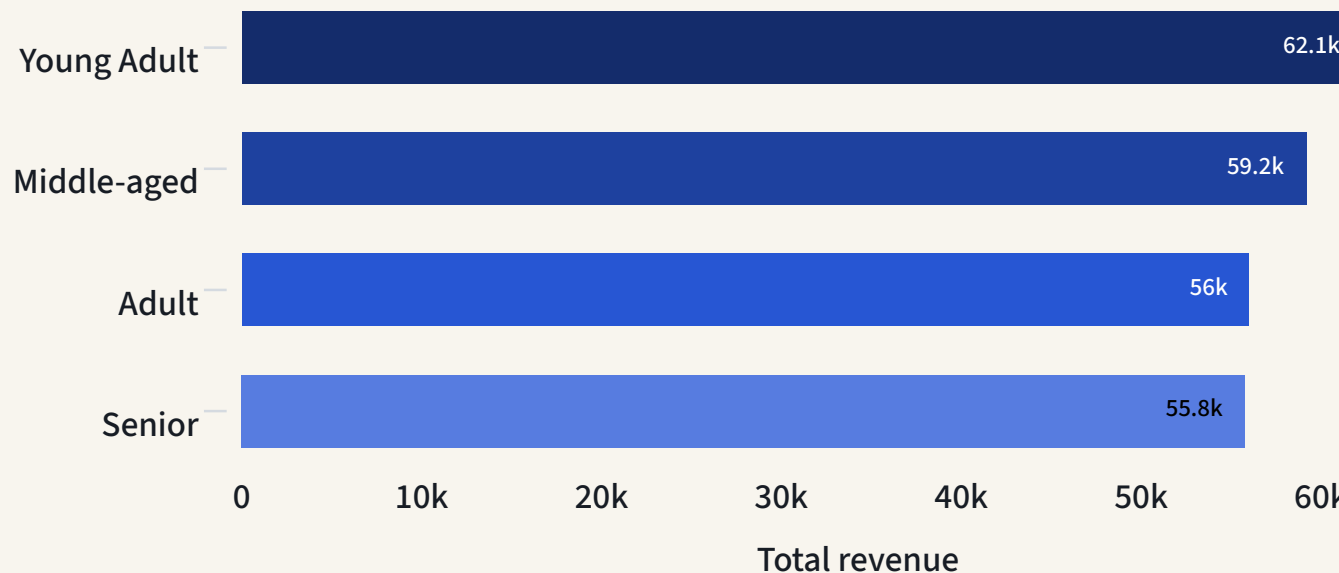
Young Adult

the largest revenue-contributing age group

Use age groups as a targeting lens, not a standalone strategy.

REVENUE BY AGE GROUP

Age group



SO WHAT

Pair age group with product, promotion, and subscription behaviour.

Five moves turn the analysis into action

01 Boost subscriptions

Add exclusive benefits and convert high-frequency buyers.

02 Build loyalty

Reward repeat buyers and move more customers into the Loyal segment.

03 Protect margin

Review discount policy on promotion-dependent products.

04 Position winners

Feature top-rated and best-selling products in campaigns.

05 Target smarter

Focus marketing on high-revenue age groups and express-shipping users.